

03 December 2025

INDUSTRIAL VISIT NEWSLETTER



Summary of the visit



We visited the UTHM Digital Gateway for an industrial exposure session. Although the building is still under development, we were able to explore their digital centre and data centre facilities. Prof. Dr. Nazri delivered a talk on UTHM's ICT Strategic Plan, Enterprise Architecture (EA), and Strategic Digitalisation Plan. He emphasized that EA involves not only the ICT department but also all other departments requiring careful planning and coordination. We also gained insights from Encik Zairil, who explained data centre operations, cooling systems using hot and cold aisles, and the technologies used in managing and maintaining the facility.

Technologies and Issues

During the visit, we were introduced to UTHM's new dedicated data centre. The centre features a modern environmental control layout using hot aisle and cold aisle containment, which helps optimise airflow and improve cooling efficiency. This design supports better energy usage and ensures that servers operate within safe temperature ranges. The team also shared their roadmap for digitalisation, which includes the migration from a traditional client-server system to a fully web-based environment carried out in phases to reduce risk and ensure smooth adoption.

In addition, we were briefed on several operational issues that commonly arise, such as authentication challenges, Wi-Fi coverage, peak-time system load during student registration, and the need for efficient data integration across platforms like LMS and internal databases. These complex operational and integration issues underscore the critical need for a guiding Enterprise Architecture (EA) framework to ensure long-term, cohesive digital growth. These discussions provided a clearer picture of how UTHM plans, manages, and improves its ICT infrastructure to support future digital growth.



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From this visit, I understand Enterprise Architecture requires cross-department planning, not just ICT. I also gained clearer understanding of data centre design, especially hot and cold aisle cooling for energy efficiency. The explanations on infrastructure management, strategic ICT planning, and real operational challenges helped me connect theory to real industry practices.

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This visit was my first time learning in detail about hot aisle and cold aisle containment, and I found it fascinating that some places already use liquid cooling. I also saw how modern tools can visualise server conditions in 3D, helping to detect issues without physical inspection. It gave me a clear understanding of Enterprise Architecture's vital role in supporting systematic digital planning in an organisations, which is a valuable insight for my future ICT knowledge.